
DVH Analysis Script Documentation

Release 2.7.0.0

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April 07, 2016

CONTENTS

1	Introduction	1
1.1	Background	1
1.2	Metrics	1
1.3	Dose Bio-Correction	2
2	Quick-Start	3
2.1	Running the Script	3
2.2	Calculating Metrics	3
2.3	Structure Volumes	4
2.4	Viewing DVHs	4
3	Metrics	6
3.1	Editing Metrics	6
3.2	Creating Metrics	7
4	Templates	10
4.1	Applying a Template	10
4.2	Creating a template	10
4.3	Editing a template	11
5	Exporting and Reporting	13
5.1	Exporting Metrics	13
5.2	Exporting DVHs	13
5.3	Bio-eval Report	14
6	Viewing Options	15
7	Troubleshooting	16

INTRODUCTION

1.1 Background

The DVH Analysis script supports bio-corrected dose evaluation with multiple dose correction schemes (e.g., LQ and LQ-L models), multiple metrics (e.g., NTCP and gEUD), and DVH viewing and exporting.

1.2 Metrics

The metrics available are as follows:

Mean Dose The mean dose given to the structure.

Min Dose The absolute minimum dose given to the structure.

Max Dose The absolute maximum dose given to the structure.

Dose to Volume The least dose that the given volume receives. For example, if $D_{20\%}(Gy) = 50$ Gy, it means that 20% of the volume receives at least 50 Gy.

Volume with Dose The volume that receives the given dose or more. For example, if $V_{70Gy}(cc) = 90$ cc, it means that 90 cc of the volume receives at least 70 Gy.

Cold Volume with Dose The coldest volume that receives the given dose or less. It is defined as $W - V_x$, where W is the total volume, V is the volume-with-dose metric, and x is the given dose. For example, if $CV_{50Gy}(\%) = 25\%$, it means that the coldest 25% of the volume receives at most 50 Gy.

Dose Complement to Volume The largest dose that the given coldest volume receives. It is defined as $D(W-x)$, where D is the dose-to-volume metric, W is the total volume, and x is the given volume of interest. For example, if $DC_{20\%}(Gy) = 20$ Gy, it means that 20% of the volume receives at most 20 Gy.

NTCP The normal tissue complication probability.

gEUD The generalized equivalent uniform dose.

1.3 Dose Bio-Correction

Bio-corrections are done on the dose grid level, not based on the Eclipse-generated DVHs in the plan or plan sum.

QUICK-START

2.1 Running the Script

To run the DVH Analysis script, first make sure that the External Beam Planning application (i.e., Eclipse) version 13.6 is running and the patient and plan(s) of interest are open. Note that only a single plan sum may be opened at once. Then:

1. Click on the **Tools** menu item, then select **Scripts...**
2. Click on the **Change Directory...** button to choose the location of the script. This location should have been provided to you.
3. Click on **Run**.

2.2 Calculating Metrics

To calculate a metric for a structure:

1. On the left side of the window, click on the structure of interest (you may click on the plus symbol or on the structure label). An empty combo box will appear under the **Metric** column on the center of the window
2. Click on the combo box and select the metric you would like to calculate (see Figure 2.1). The metrics listed are grouped by their type.

The metric calculation may take a few seconds. The result for each plan will be displayed under the appropriate column. This calculation process may be repeated for as many structures as desired. The same structure may be chosen multiple times to calculate several metrics on that structure. Metrics for the same structure are grouped together (see Figure 2.2). To calculate metrics for all structures, click on the **Add All** button below the **Structures** list.

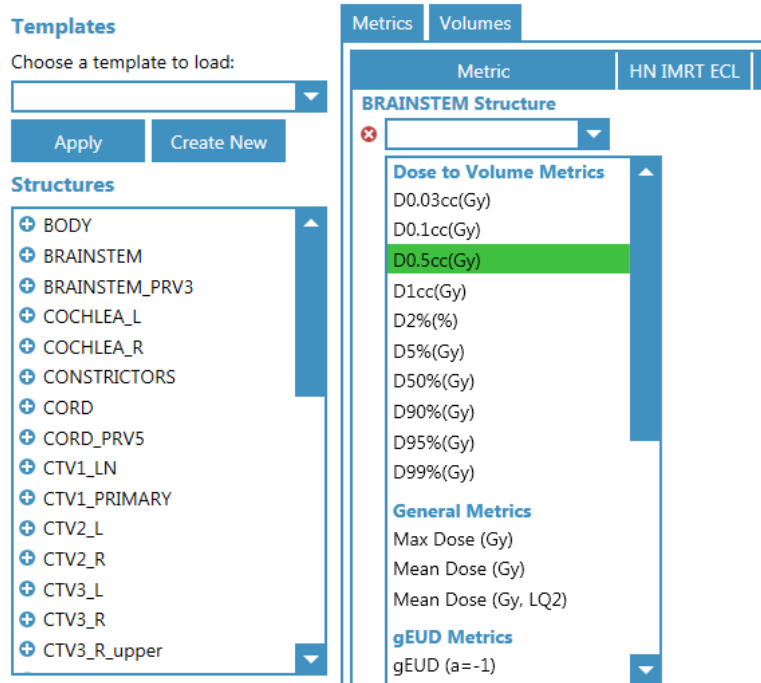


Figure 2.1: A metric for the BRAINSTEM structure is being selected.

2.3 Structure Volumes

The volume of each structure for which a metric has been calculated is reported in the **Volumes** tab (next to the **Metrics** tab). The volume is reported for each plan, as different plans may have different structure sets and the same structure in different structure sets may not be identical.

2.4 Viewing DVHs

To view the DVH curves for the structures of interest, click on the **View DVHs** button at the bottom of the window. Each structure per plan will be shown as an individual curve (see Figure 2.3). If any metrics with dose bio-correction have been calculated, the bio-corrected DVH will also be shown as an additional curve. The legend for each DVH curve will contain the name of the structure, the plan the structure belongs to, and whether the dose has been bio-corrected (e.g., **EQD2-LQ** or **EQD2-LQL**).

The DVH curves are initially shown as cumulative curves. To view the DVH curves as standard histograms, click on the **Direct** option under the **DVH Type** label. The volume unit may also be changed to relative (i.e., a percent from the total volume) by clicking on **Relative** under the **Volume** label. At the moment, the dose is always shown as absolute and cannot be changed to relative.

Metrics		Volumes		
Metric		HN IMRT ECL	HN VMAT ECL	HNSum_50Perc
BRAINSTEM Structure				
✖	D0.5cc(Gy)	36.84 Gy	40.27 Gy	38.39 Gy
✖	Max Dose (Gy)	44.18 Gy	45.38 Gy	44.41 Gy
CORD Structure				
✖	Max Dose (Gy)	43.79 Gy	42.79 Gy	43.04 Gy
✖	Mean Dose (Gy)	29.38 Gy	30.42 Gy	29.89 Gy
✖	V20Gy(%)	86.75%	87.32%	87.08%
LARYNX Structure				
✖	Mean Dose (Gy, LQ2)	12.62 Gy (LQ2)	13.14 Gy (LQ2)	12.88 Gy (LQ2)
✖	V20Gy(%)	24.56%	35.71%	34.32%
PTV56_L Structure				
✖	Mean Dose (Gy)	56.40 Gy	58.08 Gy	57.23 Gy
PTV56_R Structure				
✖	Mean Dose (Gy)	57.79 Gy	57.68 Gy	57.72 Gy
ORAL_CAVITY Structure				
✖	D0.1cc(Gy)	74.00 Gy	74.01 Gy	73.78 Gy
✖	D99%(Gy)	7.82 Gy	8.59 Gy	8.26 Gy

Figure 2.2: Results for metrics calculated for several structures, including different metrics for the same structure. Metrics per structure are calculated per plan (each column represents a plan).

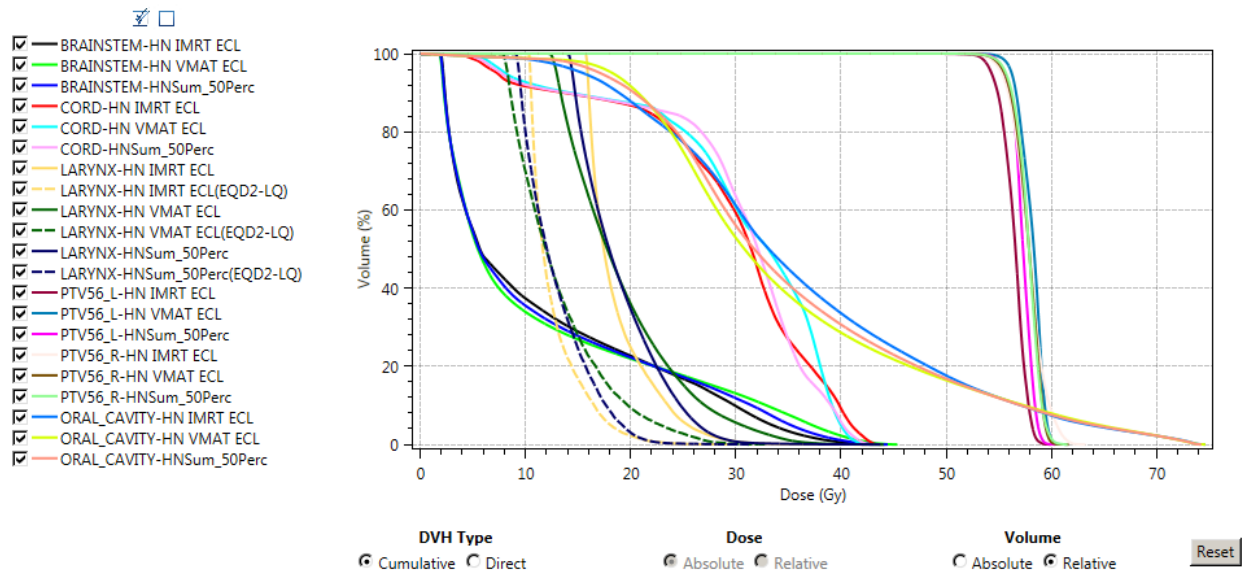


Figure 2.3: DVH curves for several structures.

METRICS

Various metrics may be calculated per structure for each opened plan. Refer to *Calculating Metrics* in Chapter 2 for details on how to calculate a metric. Any metric calculated for a structure may be deleted by clicking on the red cross symbol next to the unwanted metric. To delete all metrics for all structures, click on the **Remove All** button below the **Structures** list.

3.1 Editing Metrics

Once a metric has been calculated for a structure, it may be changed using the panel on the right side of the window (Figure 3.1). Once changed, click on the **Recalculate** button to re-calculate the metrics for the selected structure. Notice that the name of the metric displayed on the screen will have the word **[Changed]** appended to it to indicate that the metric has been changed from the original. Any changes to a metric are in effect for the selected structure only. If you want to use the original metric on that structure, click on the red cross button next to the metric to delete it, then add the structure again and choose the original metric.

Because a metric actually operates on a DVH, the properties of the metric and the DVH are separated into **Metric Properties** and **DVH Model Properties**.

Metric	HN IMRT ECL	HN VMAT ECL	HNSum_50Perc
ORAL_CAVITY Structure			
☐ D99%(Gy)	7.82 Gy	8.59 Gy	8.26 Gy
NAPE Structure			
☐ D5%(Gy)	39.67 Gy	37.76 Gy	37.53 Gy
☐ Max Dose (Gy)	52.20 Gy	52.17 Gy	49.35 Gy

Metric Properties
Metric type:
Volume: %

DVH Model Properties
DVH model type:
Dose type: ☒ Absolute ☐ Relative
Volume type: ☐ Absolute ☒ Relative

Figure 3.1: A metric may be changed temporarily using the options on the right.

3.1.1 Metric Properties

In the **Metric Properties** section, the metric type may not be changed, but any metric-specific parameters may be changed. The following table shows the parameters that may be changed for each metric type.

Metric	Parameters
Dose to volume	Volume
Volume with dose	Dose
Dose complement to volume	Volume
Cold Volume with dose	Dose
NTCP	n, m, TD50
gEUD	a

3.1.2 DVH Model Properties

In the **DVH Model Properties** section, the type of DVH model may be changed as well as any related parameters. There are three model types: Standard, LQBioDose, and LQLBioDose. For most metrics, the dose and volume types (absolute or relative) may be changed. However, for the NTCP and gEUD metrics, the dose and volume types are always absolute and cannot be changed. The dose and volume types may affect the unit of a metric parameter. For example, choosing **Relative** for the volume type in the DVH model will cause the **Volume** parameter in the **Dose to Volume** metric to be in percent rather than in cc.

The following parameters for each DVH model may be changed:

DVH Model	Parameters
Standard	(None)
LQBioDose	α/β
LQLBioDose	α/β , DT

Note that when the DVH model is changed to a bio-corrected dose (LQ or LQL), the α/β value will be automatically chosen as 2.5 for normal structures and 10 for target structures (GTV, CTV, or PTV). This value may be changed and the metric re-calculated.

The LQL bio-correction model is currently experimental. For more information on the LQ vs. LQL model, please see this link: <http://www.eyephysics.com/tdf/Models.htm>.

3.2 Creating Metrics

New metrics may be created and old ones changed using the **Metric Editor**. Click on the **Metrics...** button at the bottom of the script window to open the **Metric Editor** (Figure 3.2). On the left side of the window, the current list of available metrics are shown, grouped by type and in alphabetical order. On the right side of the window, the properties of the metric are shown. These properties are

identical to those described in the previous section, with one set of additional properties: **General Properties**. Here, the name and description of the metric may be provided. The name of the metric is generated automatically if the option **Generate name automatically** is checked.

Metric Editor

Metrics

- Dose to Volume Metrics**
 - D0.03cc(Gy)
 - D0.1cc(Gy)
 - D0.5cc(Gy)**
 - D1cc(Gy)
 - D2%(Gy)
 - D5%(Gy)
 - D50%(Gy)
 - D90%(Gy)
 - D95%(Gy)
 - D99%(Gy)
- General Metrics**
 - Max Dose (Gy)
 - Mean Dose (Gy)
 - Mean Dose (Gy, LQ2)
- gEUD Metrics**
 - gEUD (a=-1)
 - gEUD (a=-20)
- NTCP Metrics**
 - NTCP Liver Metastatic
 - NTCP Liver Primary
 - NTCP Lung
- Volume with Dose Metrics**
 - V10Gy(Gy)

General Properties

Name:

☒ Generate name automatically

Description:

Metric Properties

Metric type:

Volume: cc

DVH Model Properties

DVH model type:

Dose type: ☒ Absolute ☐ Relative

Volume type: ☒ Absolute ☐ Relative

Figure 3.2: The metric editor that is used to create or modify metrics.

To create a new metric:

1. Click on the **Add** button. A new metric will be created with the default type of **Dose to Volume**.
2. Click on the combo box next to **Metric type** to change the type of metric. Once changed, the new metric will be moved to the correct group in the list of metrics.
3. Set its properties as described in the previous section.
4. Under **General Properties**, a name and a description for the metric may be given. By default, the name of the metric is generated automatically, but a custom name may be entered

by unchecking the option **Generate name automatically**.

You may add and remove metrics as necessary. To save all your changes, click on the **OK** button. To discard all your changes, click on the **Cancel** button.

TEMPLATES

A template is a set of pre-configured metrics on a set of structures. Templates make it easy to measure the same metrics for different patients. Templates that you create are known as “Personal Templates,” for which you have the freedom to edit or delete. Templates that others have created and are shared with everyone are known as “Public Templates,” for which you cannot change.

4.1 Applying a Template

To apply an existing metric to the current patient:

1. Select a template from the combo box under **Templates**, located in the upper left corner of the script window.
2. Click on the **Apply** button.

A new window will appear showing the structures in the template as well as the structures in the opened patient (Figure 4.1). For each structure in the template, choose the matching structure in the patient (the script will attempt to make a best guess). To ignore a structure, choose (**None**) as the structure in the patient. Click on **OK** to accept the template or **Cancel** to not apply the template.

4.2 Creating a template

To create a new template:

1. Add the desired metrics for the structures of interest. The metric properties may be changed and these will be saved in the new template.
2. Click on the **Create New** button under the **Templates** section, located in the upper left corner of the script window.
3. In the new window that appears (Figure 4.2), change the name of the template from **<New template>** to a meaningful name.

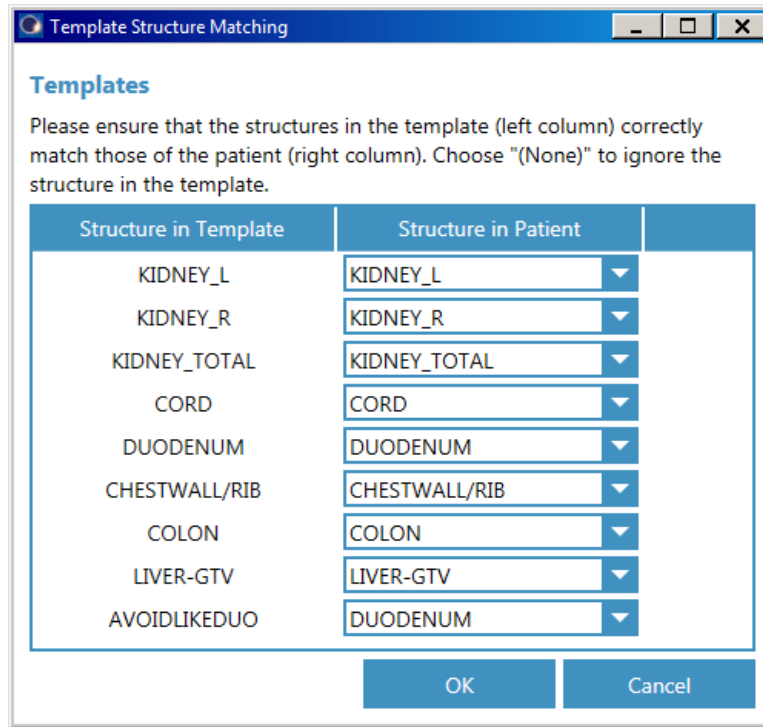


Figure 4.1: The **Template Structure Matching** window allows you to match structures in the template to structures in the patient.

4. Optionally, type in a description for the template.
5. By default, templates are private (only you can see the templates you create). If you would like to make the template available to everyone else who uses this script, check the *Make available to everyone* checkbox. You can make your template private again in the future by unchecking the checkbox.
6. Click on **OK**.
7. To cancel creating the new metric, click on the **Cancel** button.

The newly created template will appear in the combo box under **Templates** and may be used immediately.

4.3 Editing a template

For templates you created, you may edit the name, description, and whether the template is available to everyone. Currently, you may not change the individual structures and metrics related to a template. To edit a property of a template, open the **Templates** window (located at the lower right corner of the main window) and edit the desired properties.

You may create a new template based on another template by first applying the template, then

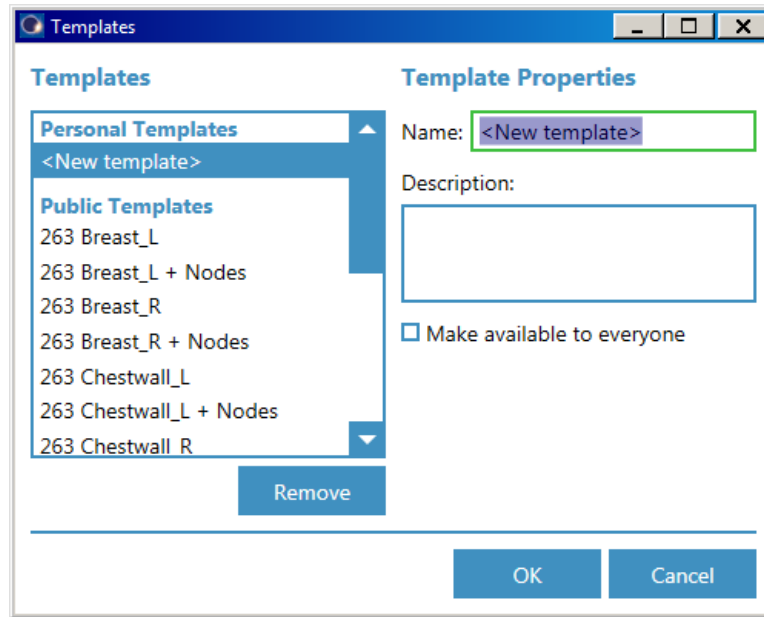


Figure 4.2: The **Templates** window lets you modify templates.

modifying the metrics (such as adding, deleting, or editing them), and clicking on the **Create New** button.

EXPORTING AND REPORTING

The metric results shown on the screen as well as the DVH curves for the chosen structures may be exported to Excel. In addition, a bio-eval report may be generated as a PDF for viewing or for uploading it to the patient's Documents in ARIA.

5.1 Exporting Metrics

To export the metric results, simply click on the **Export to Excel** button at the bottom of the script window. A new window will open showing you the export options (described below). Click on **Export** and Excel will open and create three tabbed sheets.

The first sheet (**Comparison**) contains the metrics for each structure of interest per plan (much like what is displayed on the script itself). The second sheet (**QC Analysis**) contains additional statistics on the metrics. Finally, the third sheet (**Volumes**) contains the volume of each structure (similar to what is shown in the **Volumes** tab of the script itself).

5.2 Exporting DVHs

To export the DVH curves for each structure, check the **Export DVH** option in the **Export Options** window, which is opened when you click on the **Export to Excel** button. Three options are available: **DVH Type**, **Dose Type**, and **Volume Type**. Choose the desired options for the DVHs to be exported. Note that for metrics whose dose and/or volume types cannot be changed, the DVH curves exported will be in those set units. Click on the **Export** button to export the DVH curves (as well as the metric results).

Excel will open with the tabbed sheets described above. In addition, there will be new tabbed sheets corresponding to the plans of interest. In each sheet, the DVH data for each structure is shown. If all DVH data are in the same units, they will be shown in a single plot. However, if some DVH data are in different units, more than one plot will be shown so as to keep the plot units consistent.

5.3 Bio-eval Report

To create a PDF report of biocorrected metrics, click on the **BioEval Report...** button, located at the bottom of the main window. You may change the text of any of the shown sections (physician request, physics report notes, and physician final acknowledgment). To view the PDF, click on the **Show** button, which will open the PDF for viewing. To upload the PDF to the patient's Documents in ARIA, click on the **To Documents** button. You will be notified whether the upload was successful.

VIEWING OPTIONS

The DVH Analysis script has several features that allow you to customize your viewing experience:

Column width The column widths of the metric table can be changed by dragging (to the right or left) the right border of the desired column.

Plan order The order in which plans appear can be changed by dragging and positioning the column header of the plan you would like to re-order.

Plan visibility A plan's metrics may be hidden from view by right clicking on the column header of the plan and unchecking it. Conversely, a plan can be made visible again by right clicking on any column header and checking the plan desired.

Structures list width The width of the **Structures** list can be increased or decreased by dragging the separator between the list and the metrics table. The cursor will change to a left/right arrow when it is on top of the separator.

TROUBLESHOOTING

Various messages, warnings, or errors may be shown by the DVH Analysis script in response to some expected or unexpected event. The following are the common messages or errors you may see. If you see a message or error not described here, please contact us and tell us what possibly caused the message.

Sequence contains no elements. If this error message is displayed as soon as the script is started, it may mean that there is no patient or plan open. If it is shown at a different time, please contact us.

Plan sum (plan name) contains plans with different structure sets. Metric calculations may not be correct for this plan sum.

Plan sums with different structure sets is unsupported.

PlanSum cannot have relative dose value presentation. Error occurred for plan (plan name).

Plan sums must have absolute dose type. Change the DVH model such that the dose type is **Absolute**.

Some points don't have valid doses. Perhaps the dose grid doesn't cover the whole structure.

Ensure that the dose calculation grid includes the entire structure of interest.